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clc;
close all;

% ===== Standard parameters =====
f = [800 1400 2000];
d = [3 6 9];
numW1 = [0 4];
numW2 = [0 2];
numF1 = [0 2];

% Vectors
ff = 800:50:2000;
dd = [1:0.4:5 5.5:0.5:14 14.6:0.6:20];
thinWalls = 1:20;
thickWalls = 1:5;
floors = 1:20;

% Result vectors
fLoss = zeros(3, length(ff));
dLoss = zeros(3, length(dd));
wallLoss1 = zeros(3, length(thinWalls));
wallLoss2 = zeros(3, length(thickWalls));
floorLoss = zeros(3, length(floors));
totalFLoss = zeros(9, length(ff));
totalDLoss = zeros(9, length(dd));

% Path Loss vs Frequency
for i = 1:length(ff)

    fLoss(1,i)= mwm(ff(i), d(1), 0, 0, 0);
    fLoss(2,i)= mwm(ff(i), d(2), 0, 0, 0);
    fLoss(3,i)= mwm(ff(i), d(3), 0, 0, 0);

end

figure;
plot(ff,fLoss(1,:), '-g',ff,fLoss(2,:), '-m',ff,fLoss(3,:), '-r', 'LineWidth',
    3);
title('Path Loss vs Frequency');
legend('d=3m', 'd=6m', 'd=9m', 'Location', 'southeast');
ylabel('Path Loss (dB)');
xlabel('Frequency (MHz)');
grid on;

% Path Loss vs Distance
for i = 1:length(dd)

    dLoss(1,i)= mwm(f(1), dd(i), 0, 0, 0);
    dLoss(2,i)= mwm(f(2), dd(i), 0, 0, 0);
    dLoss(3,i)= mwm(f(3), dd(i), 0, 0, 0);

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end

figure;
plot(dd,dLoss(1,:), '-g', dd,dLoss(2,:), '-m', dd,dLoss(3,:), '-r', 'LineWidth',
3);
title('Path Loss vs Distance');
legend('f=800MHz', 'f=1400MHz', 'f=2000MHz', 'Location', 'southeast');
ylabel('Path Loss (dB)');
xlabel('Distance (m)');
grid on;

% Path loss vs Number of Wooden Walls
for i = 1:length(thinWalls)

    wallLoss1(1,i)= mwm(f(1), dd(2), thinWalls(i), 0, 0);
    wallLoss1(2,i)= mwm(f(2), dd(2), thinWalls(i), 0, 0);
    wallLoss1(3,i)= mwm(f(3), dd(2), thinWalls(i), 0, 0);

end

figure;
plot(thinWalls,wallLoss1(1,:), '-g', thinWalls,wallLoss1(2,:), '-
m', thinWalls,wallLoss1(3,:), '-r', 'LineWidth', 3);
title('Path Loss vs Number of wooden walls');
legend('f=800MHz, d=6m', 'f=1400MHz, d=6m', 'f=2000MHz,
d=6m', 'Location', 'southeast');
ylabel('Path Loss (dB)');
xlabel('Number of walls');
grid on;

% Path loss vs Number of Concrete Walls
for i = 1:length(thickWalls)

    wallLoss2(1,i)= mwm(f(1), dd(2), 0, thickWalls(i), 0);
    wallLoss2(2,i)= mwm(f(2), dd(2), 0, thickWalls(i), 0);
    wallLoss2(3,i)= mwm(f(3), dd(2), 0, thickWalls(i), 0);

end

figure;
plot(thickWalls,wallLoss2(1,:), '-g', thickWalls,wallLoss2(2,:), '-
m', thickWalls,wallLoss2(3,:), '-r', 'LineWidth', 3);
title('Path Loss vs Number of concrete walls');
legend('f=800MHz, d=6m', 'f=1400MHz, d=6m', 'f=2000MHz,
d=6m', 'Location', 'southeast');
ylabel('Path Loss (dB)');
xlabel('Number of walls');
grid on;

% Path Loss vs Number of Floors
for i = 1:length(floors)

    floorLoss(1,i)= mwm(f(1), dd(2), 0, 0, floors(i));

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        floorLoss(2,i)= mwm(f(2), dd(2), 0, 0, floors(i));
        floorLoss(3,i)= mwm(f(3), dd(2), 0, 0, floors(i));

end

figure;
plot(floors,floorLoss(1,:), '-g', floors, floorLoss(2,:), '-m', floors, floorLoss(3,:), '-r', 'LineWidth', 3);
title('Path Loss vs Number of floors');
legend('f=800MHz, d=6m', 'f=1400MHz, d=6m', 'f=2000MHz, d=6m', 'Location', 'southeast');
ylabel('Path Loss (dB)');
xlabel('Number of floors');
grid on;

% Frequency tests @Building with thin walls
for i = 1:length(dd)

    % Base frequency
    totalDLoss(1,i)= mwm(f(1), dd(i), numW1(1), numW2(1), numFl(1));
    totalDLoss(2,i)= mwm(f(1), dd(i), numW1(2), numW2(1), numFl(1));
    totalDLoss(3,i)= mwm(f(1), dd(i), numW1(1), numW2(1), numFl(2));

    % Increase frequency by 75%
    totalDLoss(4,i)= mwm(f(2), dd(i), numW1(1), numW2(1), numFl(1));
    totalDLoss(5,i)= mwm(f(2), dd(i), numW1(2), numW2(1), numFl(1));
    totalDLoss(6,i)= mwm(f(2), dd(i), numW1(1), numW2(1), numFl(2));

    % Increase frequency by 150%
    totalDLoss(7,i)= mwm(f(3), dd(i), numW1(1), numW2(1), numFl(1));
    totalDLoss(8,i)= mwm(f(3), dd(i), numW1(2), numW2(1), numFl(1));
    totalDLoss(9,i)= mwm(f(3), dd(i), numW1(1), numW2(1), numFl(2));

end

figure;
plot(dd,totalDLoss(1,:), '-go', dd, totalDLoss(2,:), '-g*', dd, totalDLoss(3,:), '-gd', dd, totalDLoss(4,:), '-mo', dd, totalDLoss(5,:), '-m*', dd, totalDLoss(6,:), '-md', dd, totalDLoss(7,:), '-ro', dd, totalDLoss(8,:), '-r*', dd, totalDLoss(9,:), '-rd', 'LineWidth', 1.5);
title('Path Loss vs Distance, Building with wooden walls');
legend('f=800MHz, Line of Sight', 'f=800MHz, Multiple Walls', 'f=800MHz, Multiple floors', 'f=1400MHz, Line of Sight', 'f=1400MHz, Multiple Walls', 'f=1400MHz, Multiple Floors', 'f=2000MHz, Line of Sight', 'f=2000MHz, Multiple Walls', 'f=2000MHz, Multiple Floors', 'Location', 'southeastoutside');
ylabel('Path Loss (dB)');
xlabel('Distance (m)');
grid on;

% Frequency tests @Building with thick walls
for i = 1:length(dd)

    % Base frequency

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totalDLoss(1,i)= mwm(f(1), dd(i), numW1(1), numW2(1), numFl(1));
totalDLoss(2,i)= mwm(f(1), dd(i), numW1(1), numW2(2), numFl(1));
totalDLoss(3,i)= mwm(f(1), dd(i), numW1(1), numW2(1), numFl(2));

% Increase frequency by 75%
totalDLoss(4,i)= mwm(f(2), dd(i), numW1(1), numW2(1), numFl(1));
totalDLoss(5,i)= mwm(f(2), dd(i), numW1(1), numW2(2), numFl(1));
totalDLoss(6,i)= mwm(f(2), dd(i), numW1(1), numW2(1), numFl(2));

% Increase frequency by 150%
totalDLoss(7,i)= mwm(f(3), dd(i), numW1(1), numW2(1), numFl(1));
totalDLoss(8,i)= mwm(f(3), dd(i), numW1(1), numW2(2), numFl(1));
totalDLoss(9,i)= mwm(f(3), dd(i), numW1(1), numW2(1), numFl(2));

end

figure;
plot(dd,totalDLoss(1,:), '-go', dd,totalDLoss(2,:), '-g*', dd,totalDLoss(3,:), '-gd', dd,totalDLoss(4,:), '-mo', dd,totalDLoss(5,:), '-m*', dd,totalDLoss(6,:), '-md', dd,totalDLoss(7,:), '-ro', dd,totalDLoss(8,:), '-r*', dd,totalDLoss(9,:), '-rd', 'LineWidth', 1.5);
title('Path Loss vs Distance, Building with concrete walls');
legend('f=800MHz, Line of Sight', 'f=800MHz, Multiple Walls', 'f=800MHz, Multiple Floors', 'f=1400MHz, Line of Sight', 'f=1400MHz, Multiple Walls', 'f=1400MHz, Multiple Floors', 'f=2000MHz, Line of Sight', 'f=2000MHz, Multiple Walls', 'f=2000MHz, Multiple Floors', 'Location', 'southeastoutside');
ylabel('Path Loss (dB)');
xlabel('Distance (m)');
grid on;

% Distance tests @Building with thin walls
for i = 1:length(ff)

% Base distance between transmitter-receiver
totalFLoss(1,i)= mwm(ff(i), d(1), numW1(1), numW2(1), numFl(1));
totalFLoss(2,i)= mwm(ff(i), d(1), numW1(2), numW2(1), numFl(1));
totalFLoss(3,i)= mwm(ff(i), d(1), numW1(1), numW2(1), numFl(2));

% Increase distance by 100%
totalFLoss(4,i)= mwm(ff(i), d(2), numW1(1), numW2(1), numFl(1));
totalFLoss(5,i)= mwm(ff(i), d(2), numW1(2), numW2(1), numFl(1));
totalFLoss(6,i)= mwm(ff(i), d(2), numW1(1), numW2(1), numFl(2));

% Increase distance by 200%
totalFLoss(7,i)= mwm(ff(i), d(3), numW1(1), numW2(1), numFl(1));
totalFLoss(8,i)= mwm(ff(i), d(3), numW1(2), numW2(1), numFl(1));
totalFLoss(9,i)= mwm(ff(i), d(3), numW1(1), numW2(1), numFl(2));

end

figure;
plot(ff,totalFLoss(1,:), '-go', ff,totalFLoss(2,:), '-g*', ff,totalFLoss(3,:), '-gd', ff,totalFLoss(4,:), '-mo', ff,totalFLoss(5,:), '-m*', ff,totalFLoss(6,:), '-

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md',ff,totalFLoss(7,:), '-ro',ff,totalFLoss(8,:), '-r*',ff,totalFLoss(9,:), '-
rd', 'LineWidth', 1.5);
title('Path Loss vs Frequency, Building with wooden walls');
legend('d=3m, Line of Sight', 'd=3m, Multiple Walls', 'd=3m, Multiple
Floors', 'd=6m, Line of Sight', 'd=6m, Multiple Walls', 'd=6m, Multiple
Floors', 'd=9m, Line of Sight', 'd=9m, Multiple Walls', 'd=9m, Multiple
Floors', 'Location', 'southeastoutside');
ylabel('Path Loss (dB)');
xlabel('Frequency (MHz)');
grid on;

% Distance tests @Building with thick walls
for i = 1:length(ff)

    % Base distance between transmitter-receiver
    totalFLoss(1,i)= mwm(ff(i), d(1), numW1(1), numW2(1), numFl(1));
    totalFLoss(2,i)= mwm(ff(i), d(1), numW1(1), numW2(2), numFl(1));
    totalFLoss(3,i)= mwm(ff(i), d(1), numW1(1), numW2(1), numFl(2));

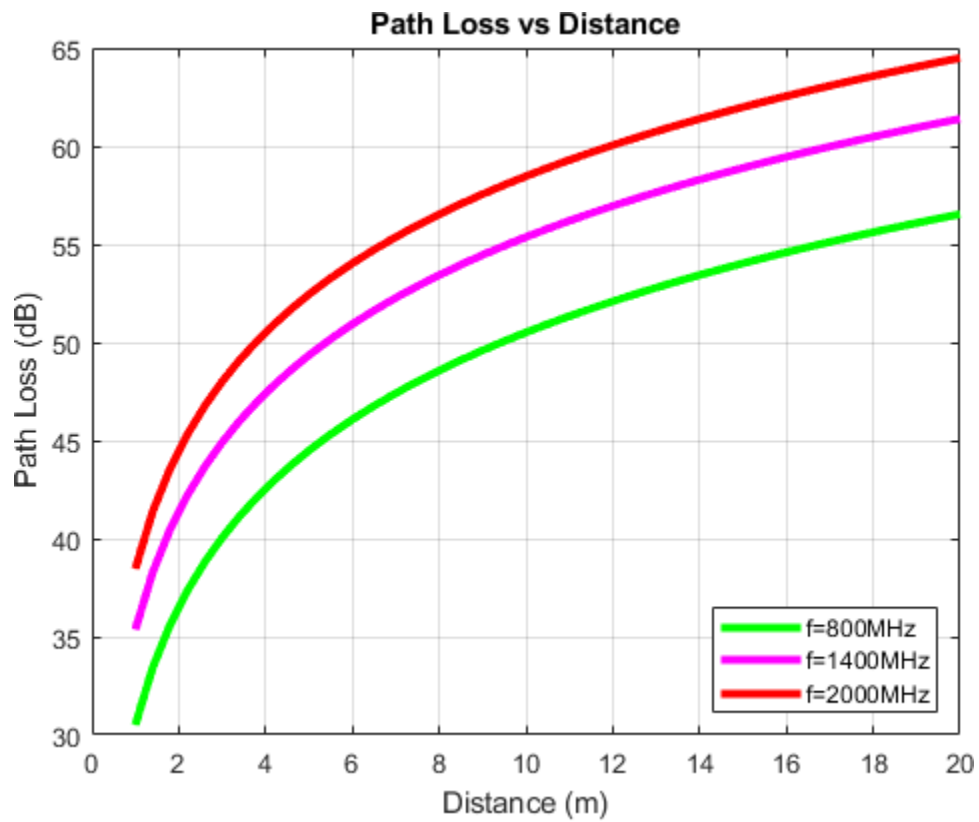
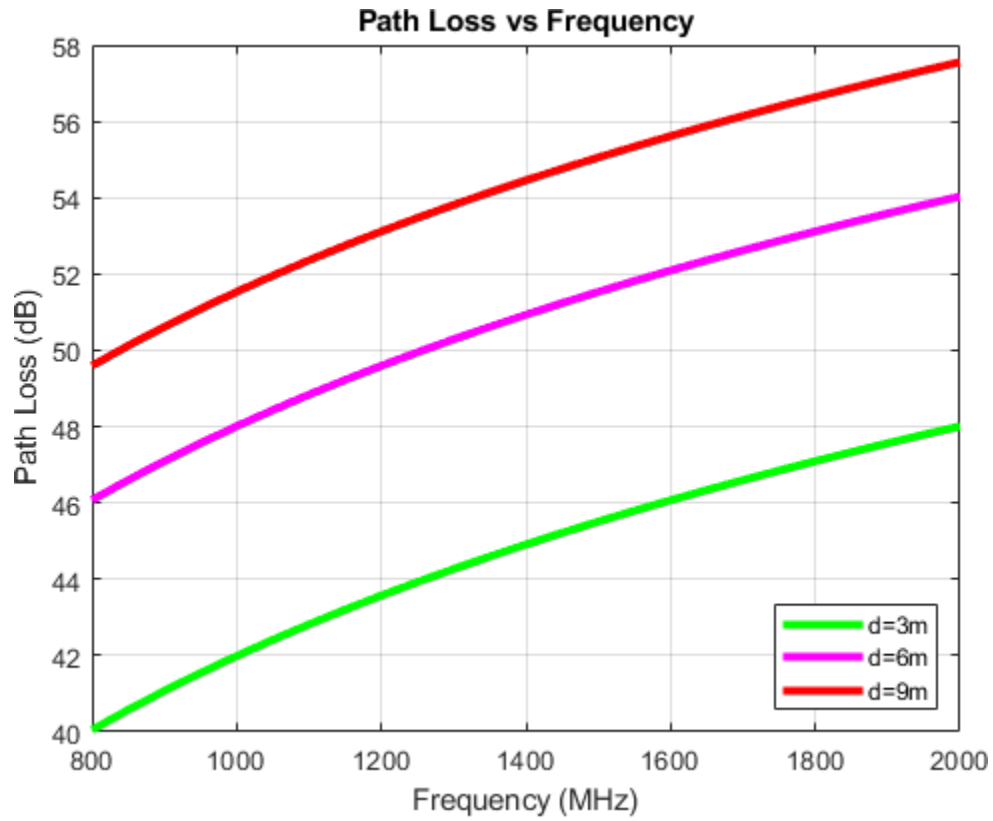
    % Increase distance by 100%
    totalFLoss(4,i)= mwm(ff(i), d(2), numW1(1), numW2(1), numFl(1));
    totalFLoss(5,i)= mwm(ff(i), d(2), numW1(1), numW2(2), numFl(1));
    totalFLoss(6,i)= mwm(ff(i), d(2), numW1(1), numW2(1), numFl(2));

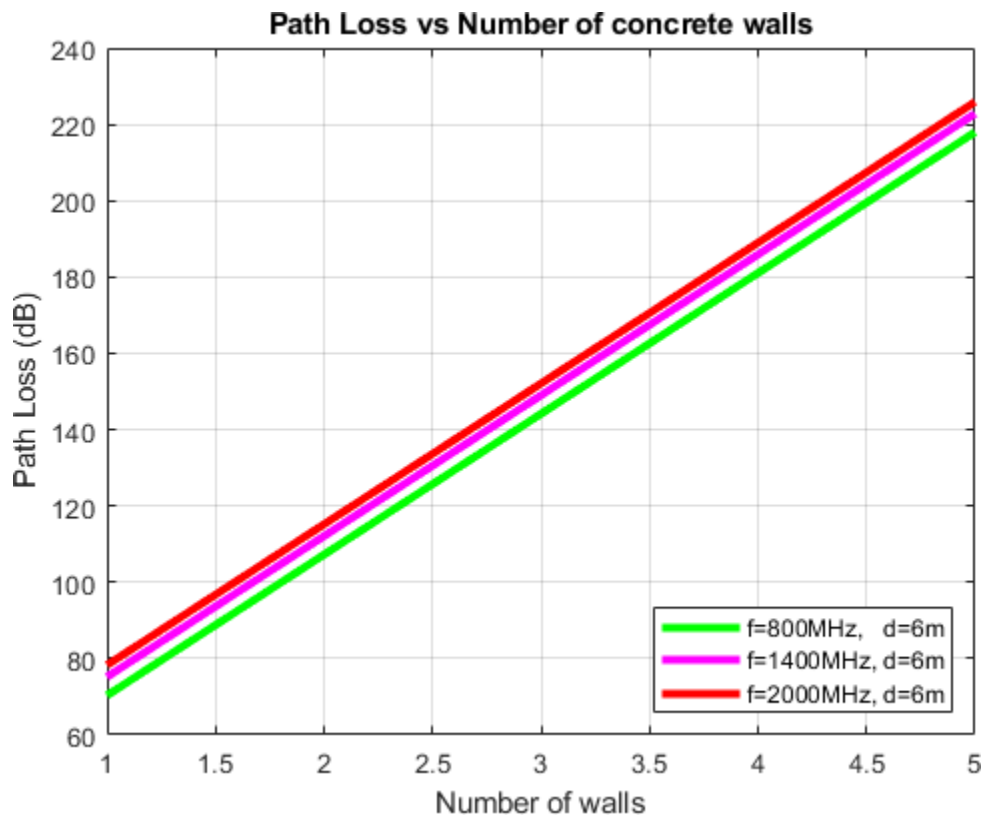
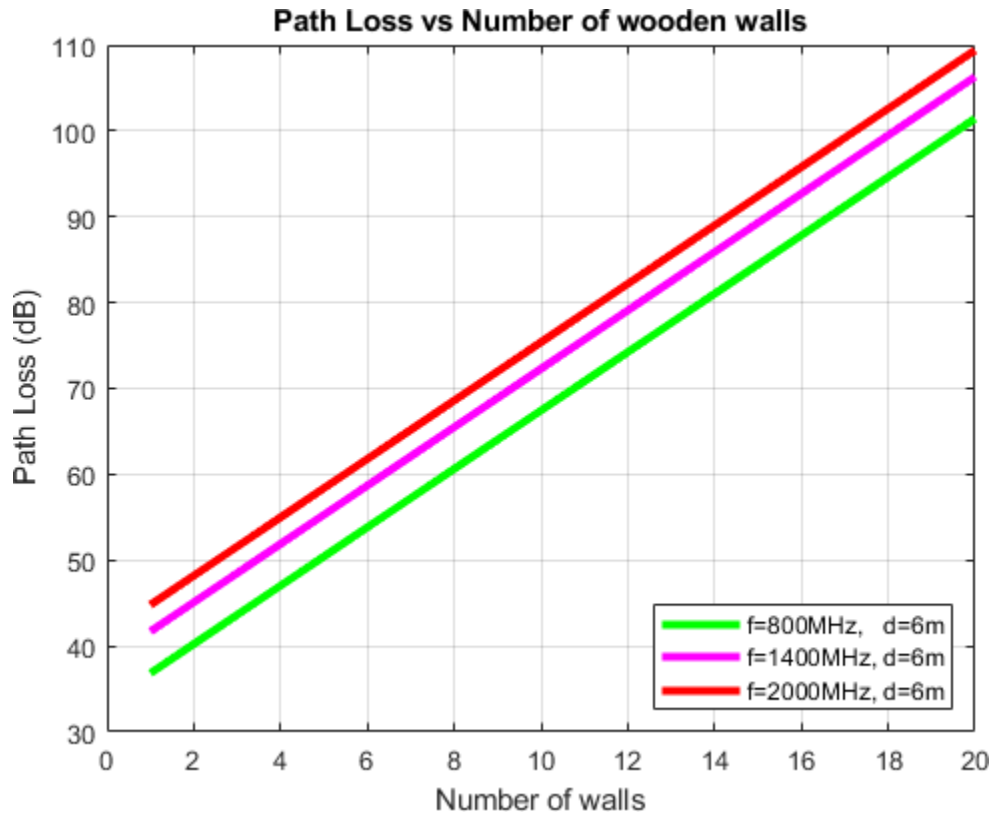
    % Increase distance by 200%
    totalFLoss(7,i)= mwm(ff(i), d(3), numW1(1), numW2(1), numFl(1));
    totalFLoss(8,i)= mwm(ff(i), d(3), numW1(1), numW2(2), numFl(1));
    totalFLoss(9,i)= mwm(ff(i), d(3), numW1(1), numW2(1), numFl(2));

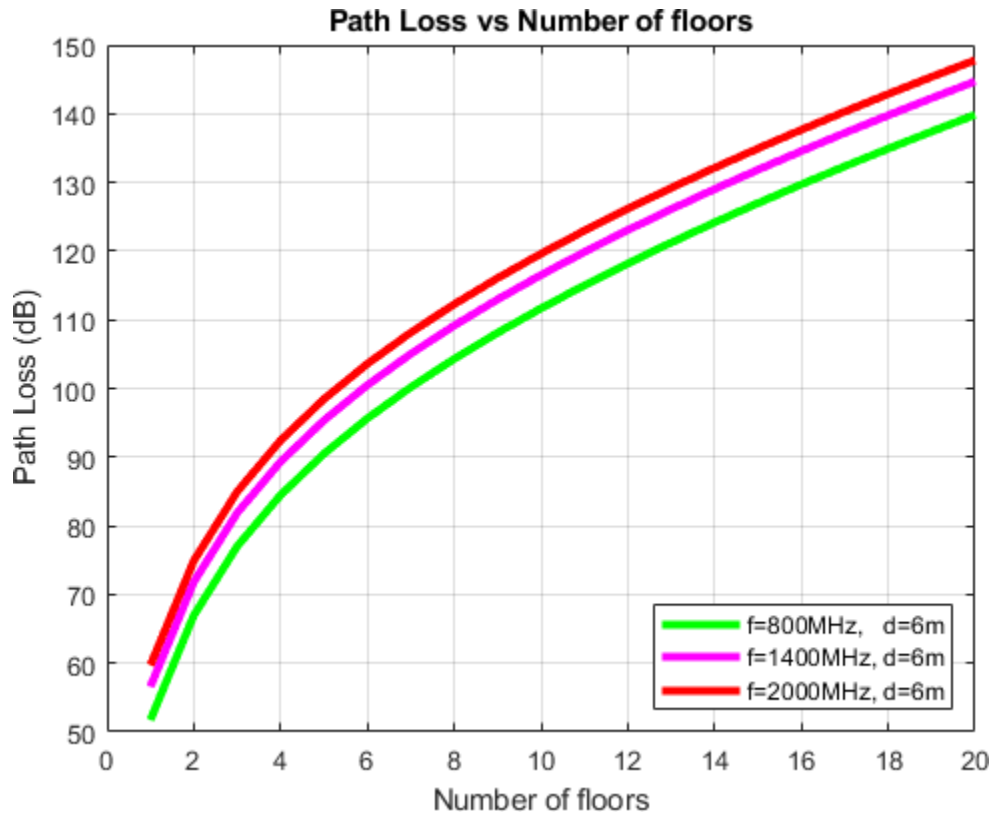
end

figure;
plot(ff,totalFLoss(1,:), '-go',ff,totalFLoss(2,:), '-g*',ff,totalFLoss(3,:), '-
gd',ff,totalFLoss(4,:), '-mo',ff,totalFLoss(5,:), '-m*',ff,totalFLoss(6,:), '-
md',ff,totalFLoss(7,:), '-ro',ff,totalFLoss(8,:), '-r*',ff,totalFLoss(9,:), '-
rd', 'LineWidth', 1.5);
title('Path Loss vs Frequency, Building with concrete walls');
legend('d=3m, Line of Sight', 'd=3m, Multiple Walls', 'd=3m, Multiple
Floors', 'd=6m, Line of Sight', 'd=6m, Multiple Walls', 'd=6m, Multiple
Floors', 'd=9m, Line of Sight', 'd=9m, Multiple Walls', 'd=9m, Multiple
Floors', 'Location', 'southeastoutside');
ylabel('Path Loss (dB)');
xlabel('Frequency (MHz)');
grid on;

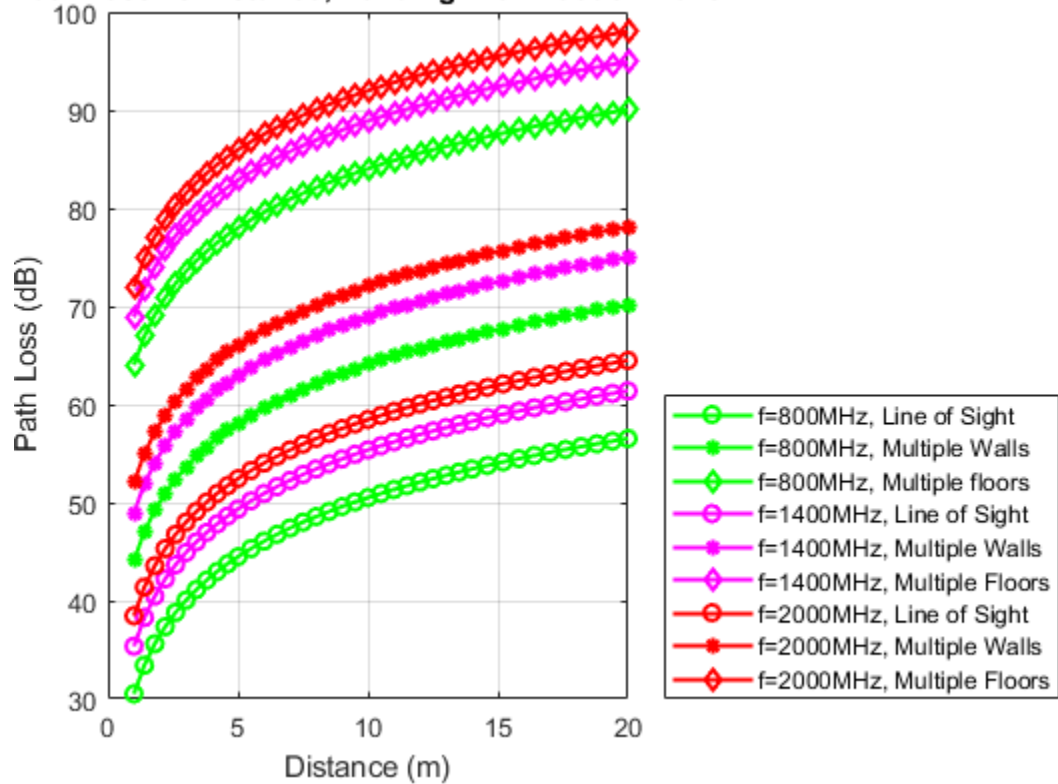
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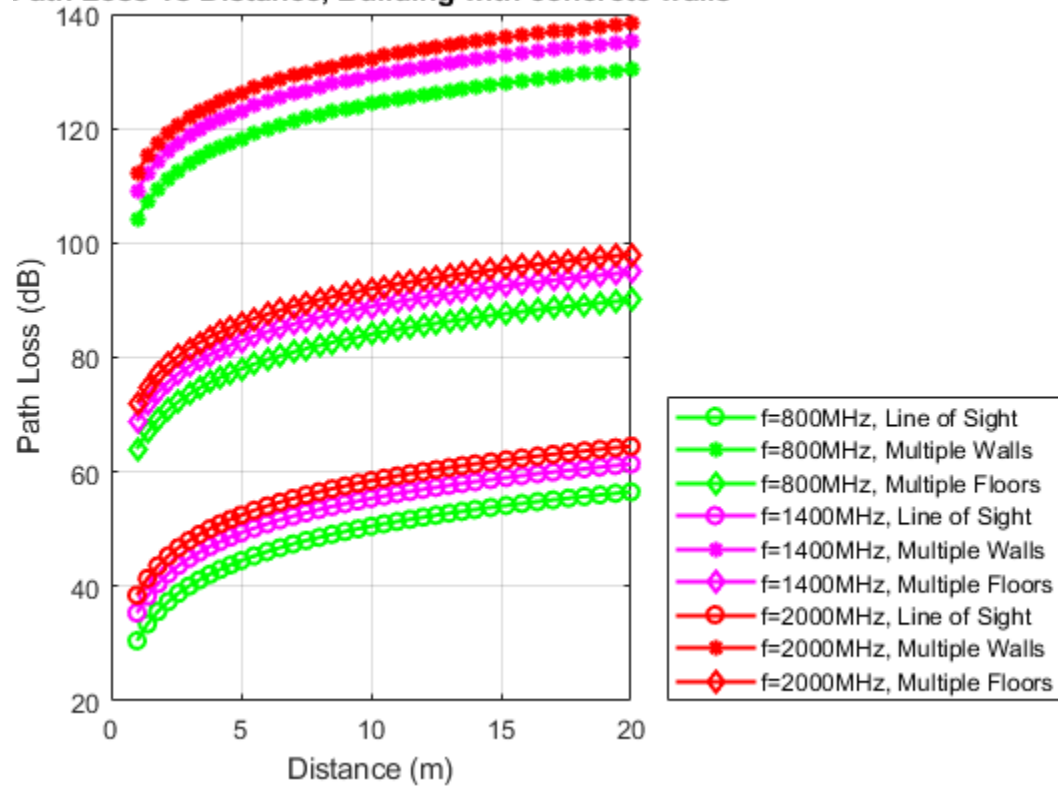




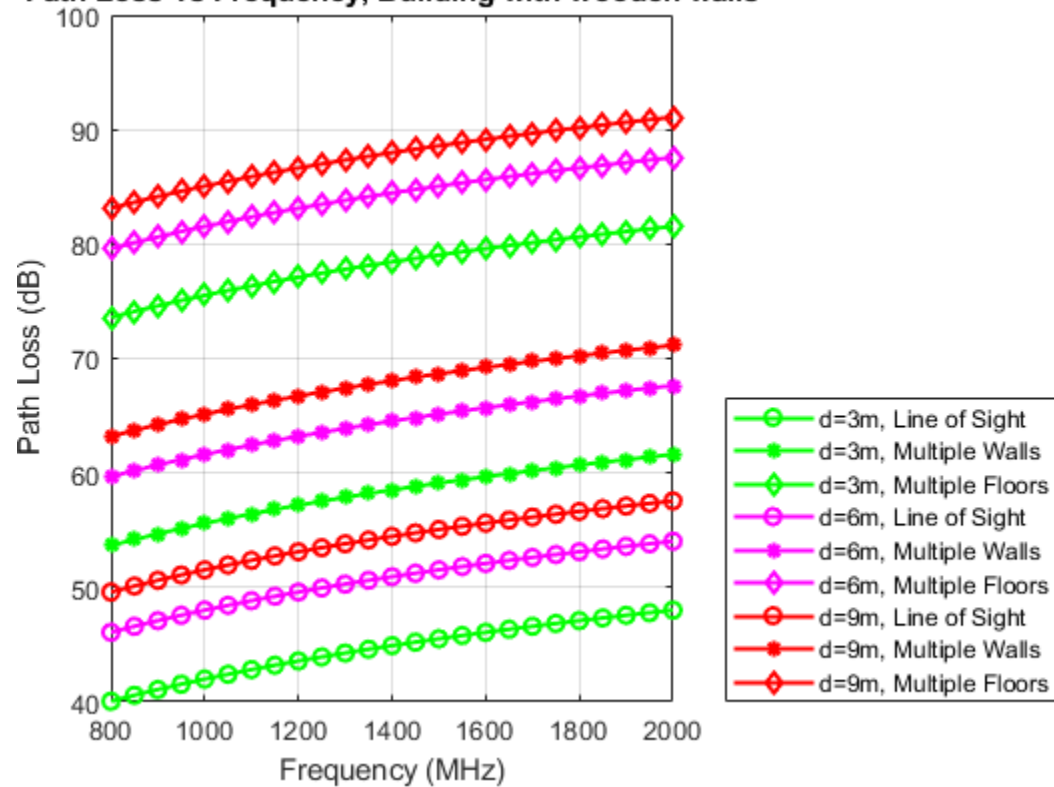
Path Loss vs Distance, Building with wooden walls



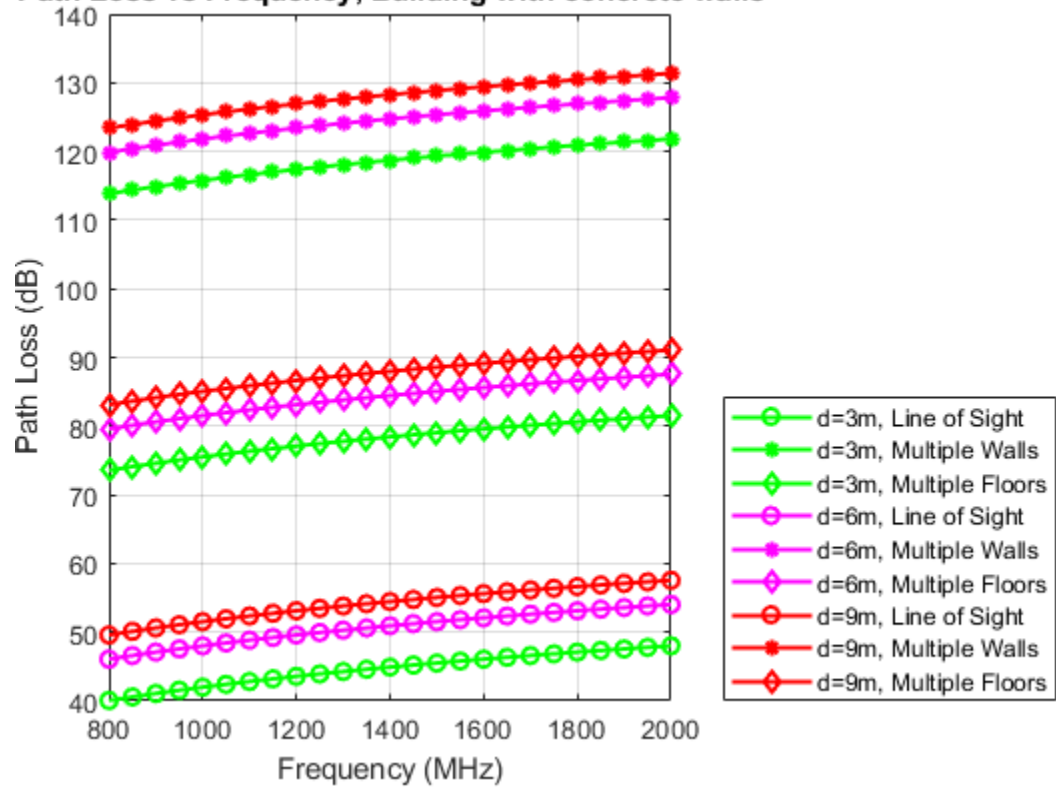
Path Loss vs Distance, Building with concrete walls



Path Loss vs Frequency, Building with wooden walls



Path Loss vs Frequency, Building with concrete walls



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